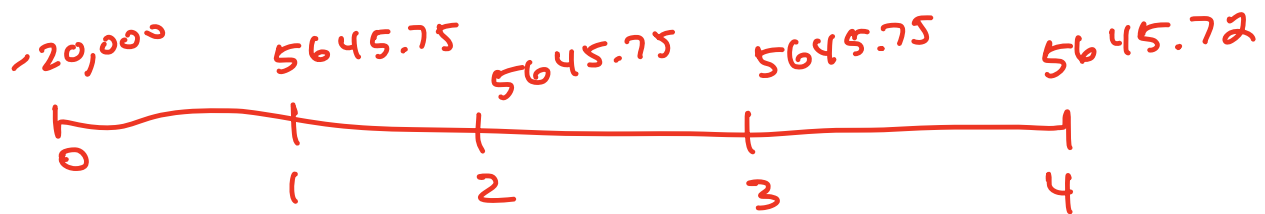


Chapter 5: In-Class Exercises

5.2

Problem: Jared borrows \$20,000 at an annual effective discount rate of 4.8%. The loan is to be repaid by annual end-of-year payments for four years, the first three of which are for the amount \$5,645.75 and the time 4 payment is for \$5,645.72. Make an amortization schedule for this loan.

$$d = .048 \Rightarrow i = \frac{.048}{.952} = .050420168$$



t	PMT	Int.	Pri.	OLB
0	-	-	-	20,000
1	5645.75	1008.40	4637.35	15362.65
2	5645.75	774.59	4871.16	10491.49
3	5645.75	528.98	5116.77	5374.72
4	5645.72	271.00	5374.72	0

5.2

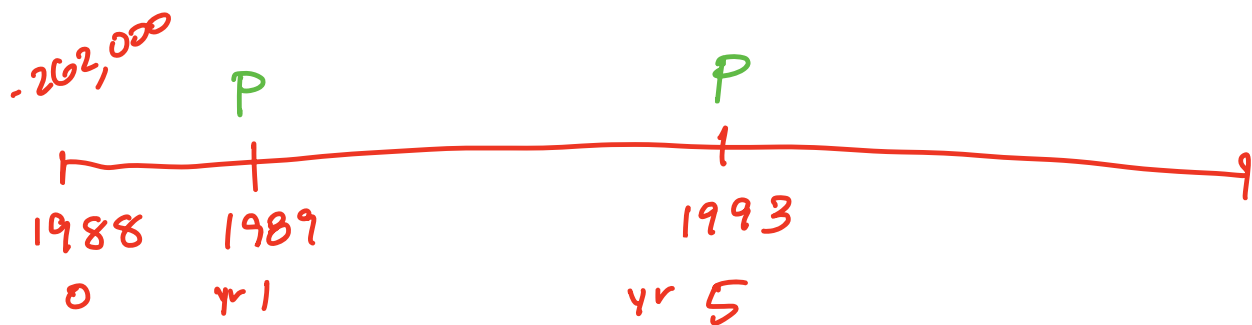
Problem: On May 1, 1988, the Ramakrishnas purchased a home for \$308,000. Their down payment was \$46,000, and the remaining \$262,000 was financed with a standard fifteen-year amortized loan at a nominal interest rate of 5.55% convertible monthly. How much interest did they pay in 1993, assuming that their payments have all been on the first of the month, the first payment was on June 1, 1988, and that there have been no additional payments beyond the scheduled ones?

Solution. $308,000 - 46,000 = 262,000$

$$15 \times 12 = 180 \text{ pmts}$$

$$i^{(12)} = .0555$$

$$i^{(12)}/12 = .004625$$



$$P a_{\overline{180}|.004625} = 262,000 \Rightarrow P = 2147.72$$

Amount of interest pmt $\underbrace{55 - 67}_{12 \text{ pmts}}$

$$= 10,922.22$$

5.4

Problem: Adam takes out a \$20,000 loan for ten years at a level annual effective interest rate of 5%. At the end of each year, he pays \$2,000 in principal, which is $\frac{1}{10}$ th of the loan amount, along with the interest due. Find a formula for P_t .

solution. $i = 5\%$ $L = 20,000$

end of 1st year : $20000(.05) = 1000$ ↖ interest
 $+ 2000$ ↖ princ.
 $P_1 = 3000$

end of 2nd year $B_1 = 18,000$

$$: 18,000(.05) + 2000 = 2900$$

In general $B_t = 20,000 - 2000t$

so

$$20,000 - 2000t = (20,000 - 2000(t-1))(1.05) - P_t$$

$$\begin{aligned} P_t &= 21,000 - 2100(t-1) - 20,000 + 2000t \\ &= 3100 - 100t \end{aligned}$$

5.4 **Problem:** Beatrice takes out a \$20,000 loan for ten years at a level annual effective interest rate of 5%. Her original payment is P and then her payments increase by \$200 each year. Find P .

Solution. If we know pmts are in Arith. Sequence we can leverage the P.V.

$$\begin{aligned}
 20,000 &= (I_{P,200} a)_{\overline{10}|.05} \\
 &= P a_{\overline{10}|.05} + \frac{200}{.05} \left(a_{\overline{10}|.05} - \frac{10}{1.05^{10}} \right) \\
 P &= 1770.274 \rightarrow 1770.28
 \end{aligned}$$

5.4 **Problem:** Cedric also takes out a \$20,000 loan for ten years at a level annual effective interest rate of 5%. His payments grow by 20% each year. Find the amount of his first payment.

solution.

$$20,000 = P \left(\frac{1 - \left(\frac{1.2}{1.05} \right)^{10}}{.05 - .2} \right) \Rightarrow P = 1070.97$$

5.4

Problem: Daphne also takes out a \$20,000 loan at a level annual effective interest rate of 5%. She makes annual payments that are four times the interest due until she can make a final end-of-year payment of no more than \$2,500 to pay off the loan. When does she make this final payment, and what is its amount?

$$P_1 = 20000 (.05)(4) = 4000$$

$$\begin{aligned} B_1 &= 20,000(1.05) - P_1 \\ &= 20,000(1.05) - 20,000(.05)(4) \\ &= 20,000(1 - 3(.05)) = 17000 \end{aligned}$$

$$P_2 = 20,000(1 - 3(.05))(.05)(4) = 3400$$

$$B_2 = 20,000(1 - 3(.05))(1.05) - P_2 = 14450$$

$$P_3 = 14450(.05)(4) = 2890$$

$$B_3 = 14450(1.05) - 2890 = 12282.5$$

$$B_t = (1.05)B_{t-1} - P_t \quad \text{and} \quad P_t = .2(B_{t-1})$$

$$\begin{aligned} B_t &= (1.05)B_{t-1} - .2(B_{t-1}) \\ &= .85B_{t-1} \end{aligned}$$

$$B_0 = 20,000, \quad B_1 = .85(20,000), \quad B_2 = .85^2(20,000)$$

$$B_t = 20,000(.85)^t$$

we want $B_t(1.05) < 2500$

$$20,000(.95)^t(1.05) < 2500$$

$$\Rightarrow t = 13.0952 \rightarrow t = 14$$

So in the 15th pmt: $B_{14}(1.05) = 2158.16$